IIT ROORKEE

FREE ONLINE EDUCATION swayam शिक्षित भारत, उन्नत भारत

NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 35: LCA-Case Study

By-
Prof. Bhola Ram Gurjar
Dept. of Civil Engineering



Contents

- LCA of biomass-based energy systems: A case study for Denmark
  - Study area
  - Background & need of the study
  - Methodology for LCA process adopted in the study
  - Energy scenarios in Denmark
  - Measures for managing energy demand
  - Impacts considerations
  - Comparison of fuels
  - Study conclusions
- Conclusion

Study area: Denmark

- Denmark is situated in Northern-Europe.
- Its capital city Copenhagen is known for its cycle-friendly infrastructure.
- It has high-income economy.



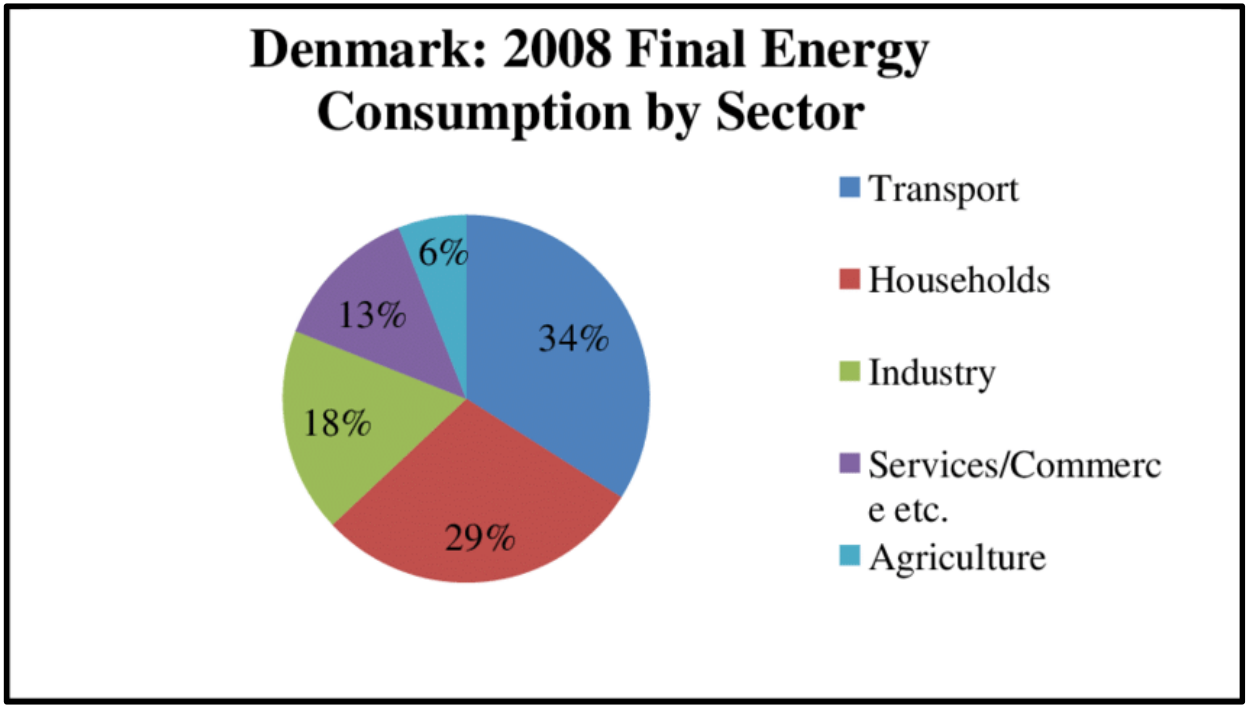
Source: (https://www.bbc.com/news/world-europe-17929661)

Background of study

- Denmark targeted to reach a 100% renewable energy system in 2050 primarily based on wind power and biomass energy but also involving significant decreases in the national energy demand

- 100% renewable energy systems can only realistically be achieved through
  - Significant reductions in energy demand,
  - Increased efficiencies of fuel conversion technologies
  - Higher shares of wind power (e.g., up to 50%)
  - Replacement of fossil fuels with biomass resource
  - Integration of the transport sector into the energy system, e.g., through establishment of electric vehicles

Sector-wise energy demand of Denmark: 2008



- Transport sector is the highest energy demand sector in Denmark.
- Households and industry are followed by it with 29 & 18% respectively.

Source: IEA, 2008

Energy sources of Denmark- 2008



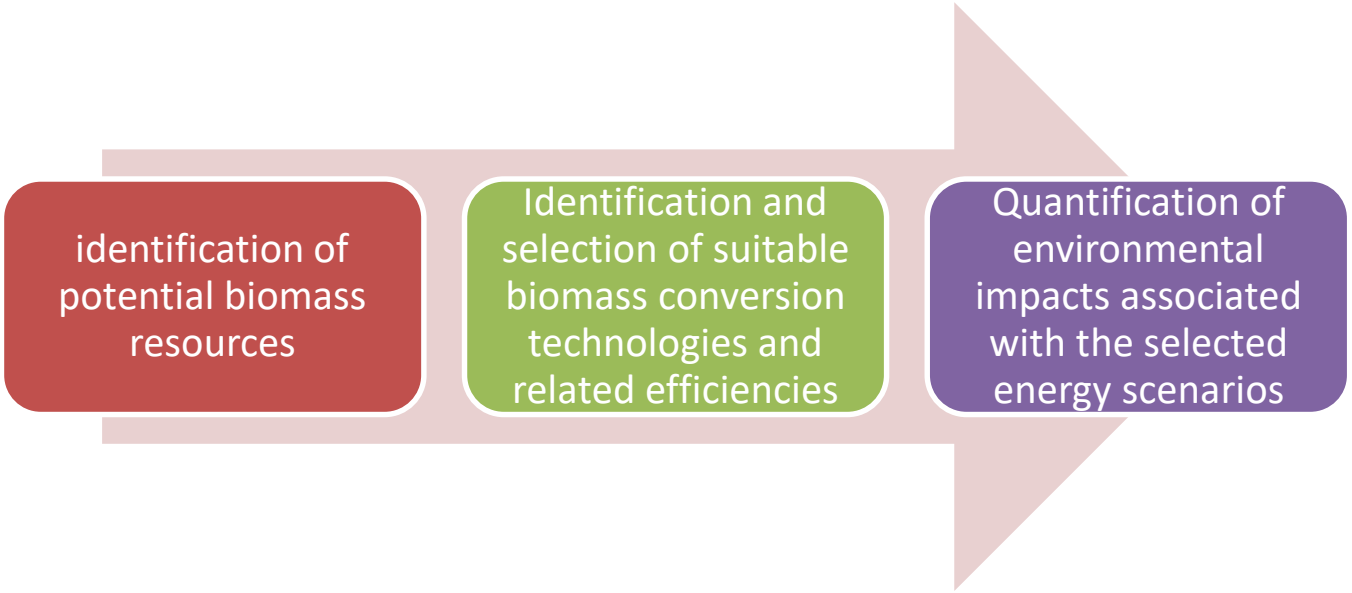
- In 2008, the share of fossil fuels in the energy system corresponded to about 84% (of the primary energy supply).
- Only 16% of the energy supply was based on renewable sources such as biomass, solar energy and waste resources
- The share of oil corresponded to 39%.

Image source: moderndiplomacy.eu

Scope of study

- Overview of the environmental consequences of changing energy production in the future needed.
- All upstream, direct and downstream emissions need to be accounted through lifecycle perspective.
- Quantification of environmental impacts associated with potential future energy scenarios for Denmark in 2030 and 2050.
- Future scenarios compared with the existing energy system in the reference year 2008.

Objectives of study



Methodology of Study: Defining the goal

- The goal of the LCA was to assess the environmental impacts related to more sustainable energy scenarios in Denmark.
- Selection of a range of potentially future energy scenarios based on a mix of residual agricultural resources, energy crops, wind and other renewables and to the extent needed including fossil fuels.
- Comparison of all future energy scenarios with a reference representing the Danish energy system in 2008.

Methodology of Study: Functional unit

- The service defined by the LCA functional unit was ‘‘meeting the energy demand (electricity, heat and fuels) required in Denmark”
- For years 2008, 2030 and 2050
- This service does not describe the amount of energy provided, but it is defined to ensure that the future living standards are same as the 2008 baseline

Methodology of Study: Scope set-up

- LCA modelling for Primary Energy Supply (PES)
- Normalized result with PES for the individual years (2008, 2030 & 2050) to allow comparison (to the intensity of the environmental impact)
- The environmental impacts were quantified with a time horizon of 100 years.

Methodology of Study: Assessment approach

- Selection of potential energy scenarios based on available national energy strategies and political targets (e.g. regarding shares of wind power, CO_2 emission reductions, etc.)
- Selection of relevant conversion technologies and collection of associated technology data for the LCA
- Balancing energy supply and demand for each scenario to provide the necessary input for the LCA
- Impact assessment and discussion of results

Methodology of Study: Impact Assessment Category

- Global warming as CO_2 eq. emission
- Impacts related to land occupation included in the assessment
- Acidification, the results were expressed as area of unprotected ecosystem per unit energy
- Aquatic eutrophication, two sub-categories were used : aquatic eutrophication (nitrogen) & aquatic eutrophication (phosphorous).

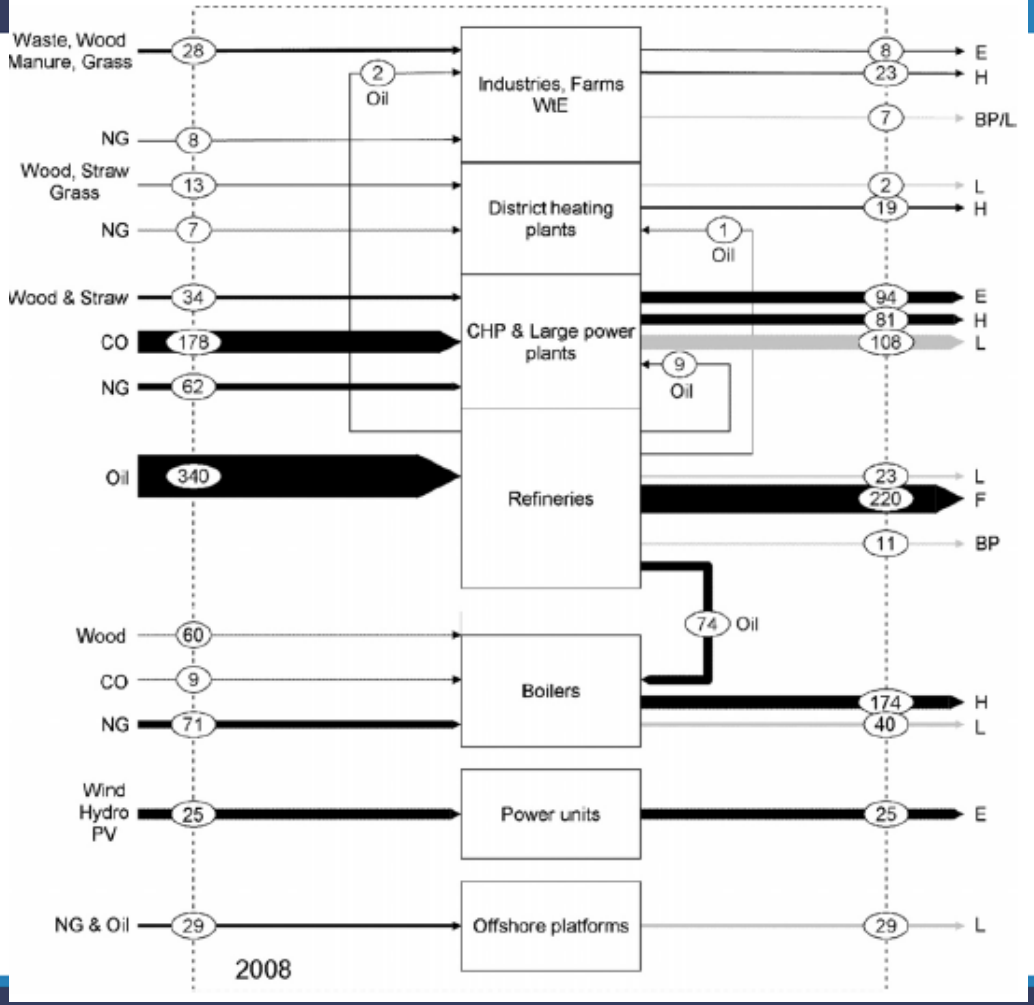
Energy Scenarios

Five different energy scenarios were assessed

- Year 2008 (Baseline)
- Year 2030
- Year 2050 - CSV (conservative scenario),
- Year 2050- RME (Bio-diesel as Rapeseed Methyl Ester (RME); totally produced from rapeseed)
- Year 2050 – BtL (Biomass-to-Liquid and Fischer–Tropsch technology, except for the share of RME corresponding to the amount produced today in DK)

The last three scenarios represent different potential alternatives for transport fuel production in year 2050.

2008 Energy Scenario

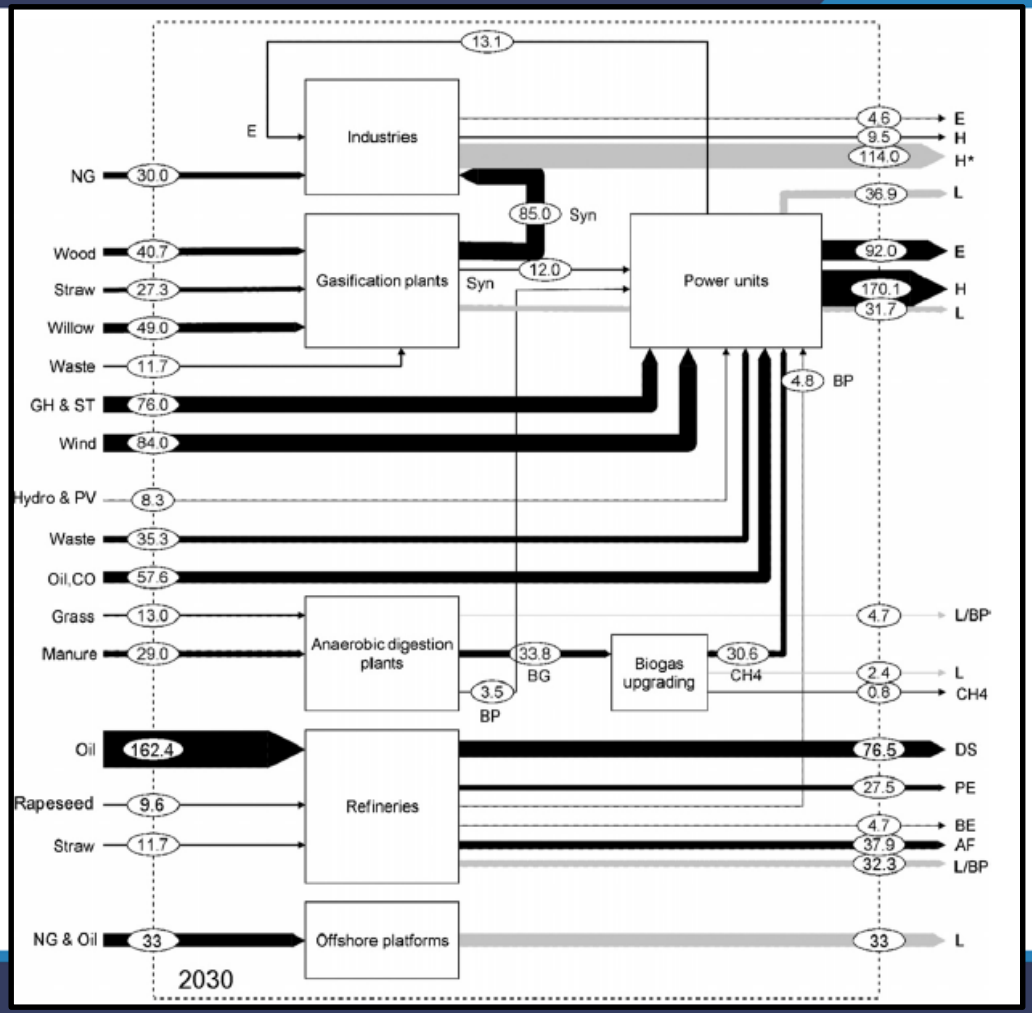


Energy demand in PJ.

One PJ= 10 to power 15

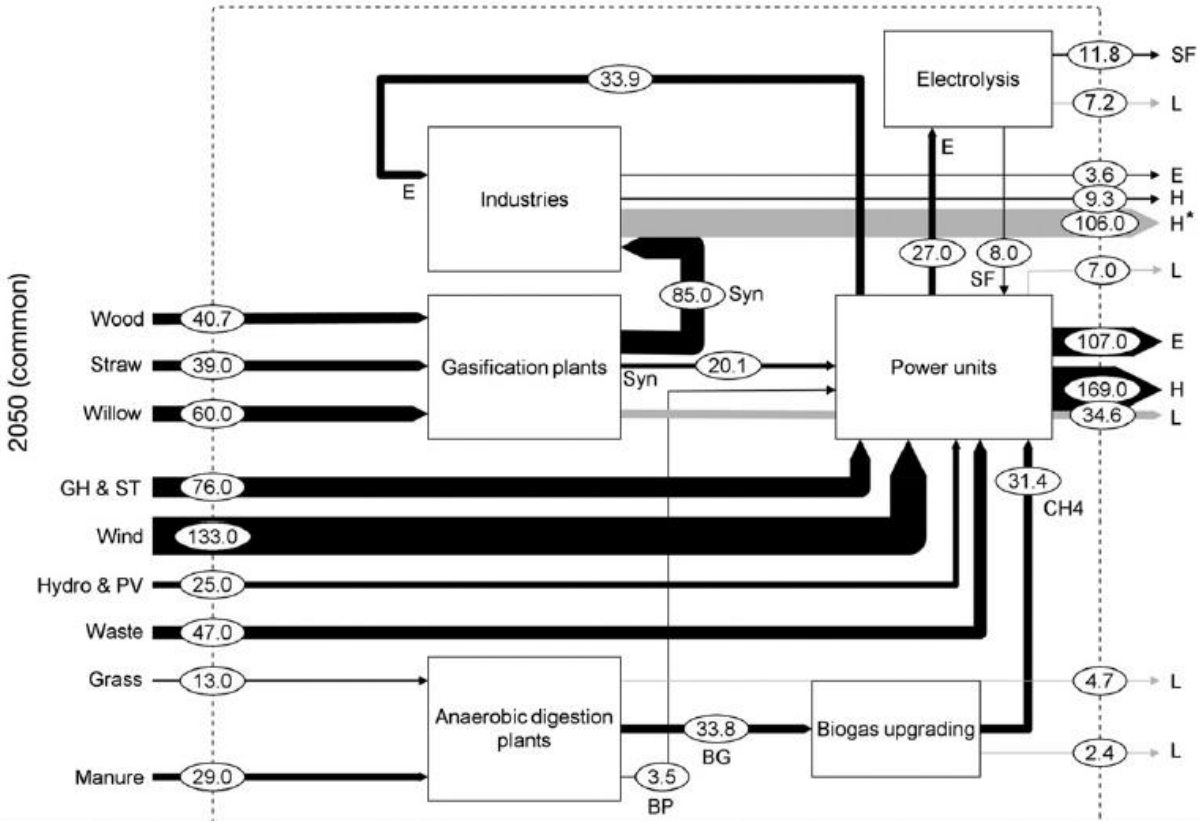
PJ- Peta Joule
NG: natural gas
CO: coal
PV: photovoltaic
Hydro: hydropower
E: electricity
H: heat
L: losses
BP: byproducts
F: fuel.

2030 Energy scenario



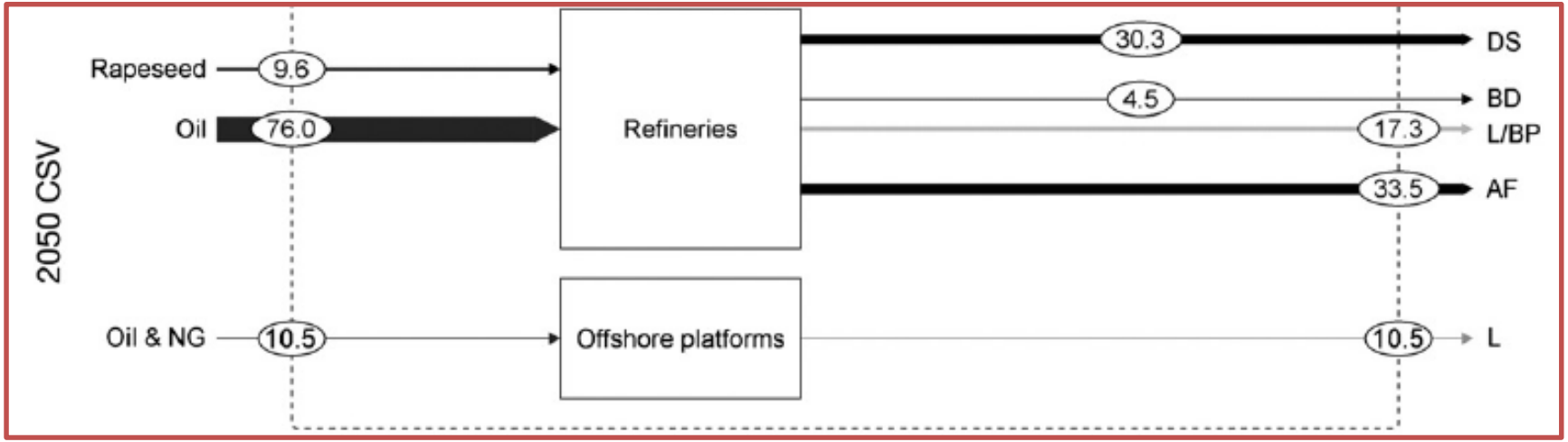
GH: geothermal heating, ST: solar heating, PV: photovoltaic, Hydro: hydropower, CO: coal, Syn: syngas, BG: biogas, BP: byproducts, E: electricity, H: heat, H: process heat (industry), L: losses, PE: petrol, DS: diesel (including 4.5 PJ of RME-biodiesel), BE: bioethanol, AF: aviation fuel.

2050 Energy Demand : Common for all 3-alternative scenario



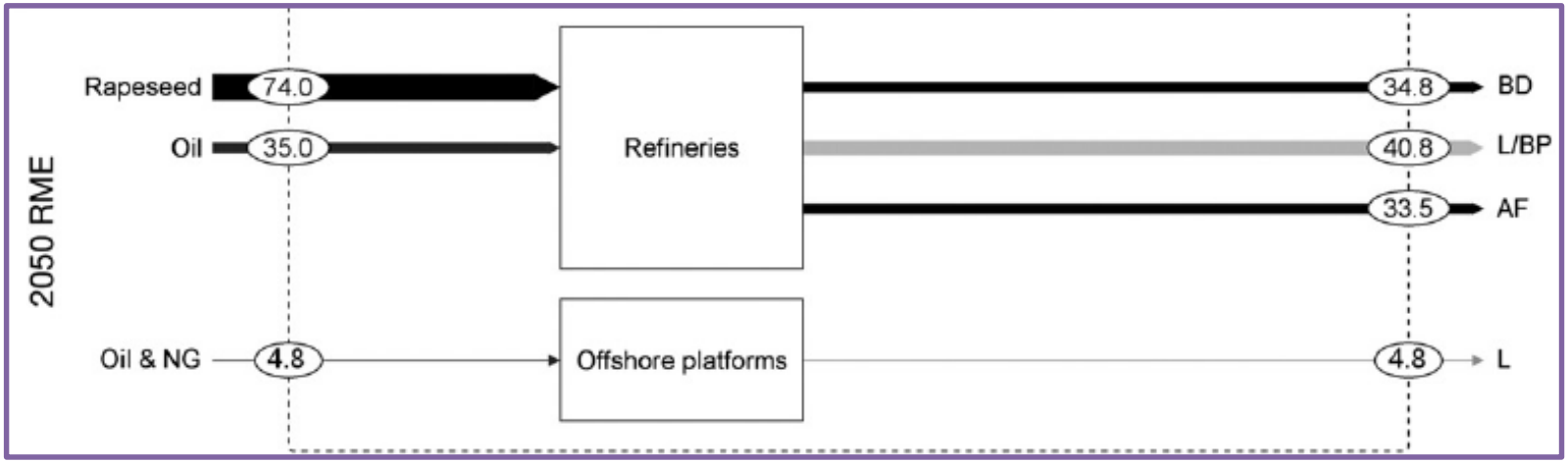
GH: geothermal heating, ST: solar heating, PV: photovoltaic, Hydro: hydropower, NG: natural gas, Syn: syngas, BG: biogas, BP: byproducts, E: electricity, H: heat, H: process heat (industry), L: losses, DS: diesel, BD: biodiesel, AF: aviation fuel, SF: synthetic fuel.

2050 Energy Demand : CSV



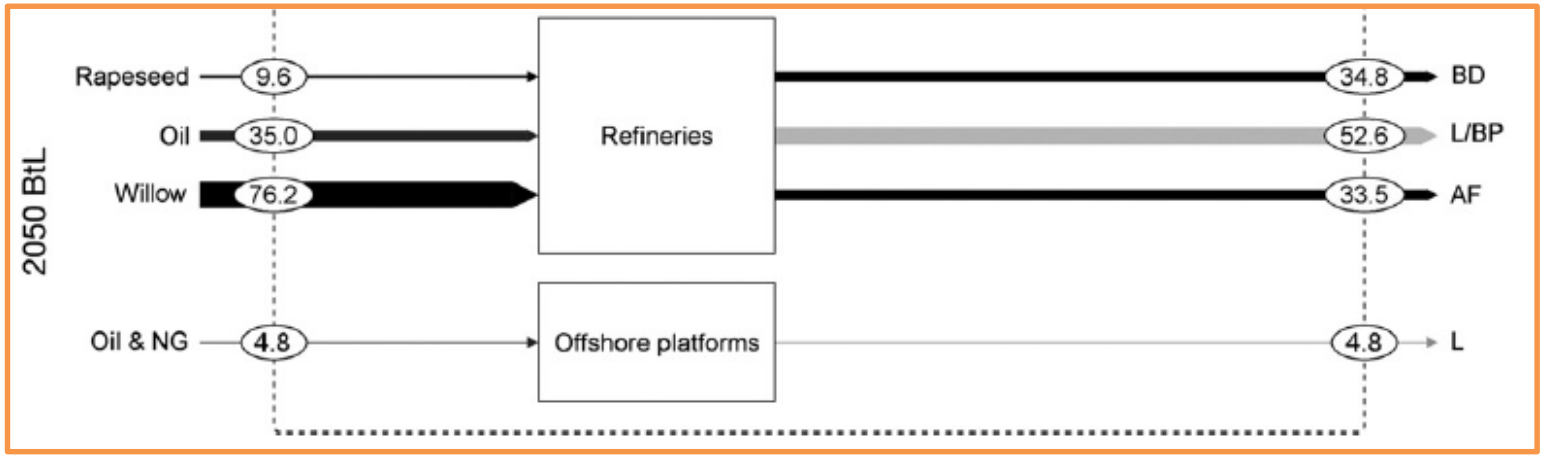
NG: natural gas, BG: biogas, BP: byproducts, E: electricity, H: heat, H: process heat (industry), L: losses, DS: diesel, BD: biodiesel, AF: aviation fuel, SF: synthetic fuel.

2050 Energy Demand : RME



BG: biogas, BP: byproducts, E: electricity, H: heat, H: process heat (industry), L: losses, DS: diesel, BD: biodiesel, AF: aviation fuel, SF: synthetic fuel.

2050 Energy Demand : BtL



BG: biogas, BP: byproducts, E: electricity, H: heat, H: process heat (industry), L: losses, DS: diesel, BD: biodiesel, AF: aviation fuel, SF: synthetic fuel.

Measures for energy demand: Electricity Production

2030	2050	Effects	Policy change
• Introduction of SOFC CHPs (33% of total CHP plants) • Geothermal plants are installed reaching a total production of about 4 TWh (15% of demand in big cities)	Introduction of SOFC CHPs (100% of total CHP plants)	Reduction of primary energy supply from fossil fuel.	• Decommission of old power plants. • Integration of geothermal with incineration plants

SOFC: Solid Oxide Full Cell
CHP: Combined Heat & Power

Measures for energy demand: Land Transport 1/2

2030	2050
The expected growth in passenger transport (18%) is assumed covered equally by train transport and by 'avoiding transport'	Growth in passenger transport (compared to 2030 value) is 'avoided'
Bicycling and walking meet 10% of transport demand (5% in 2008)	Vehicles meet 50% of the transport demand. Railroads meet 40% of the transport demand.
Vehicles meet 60% of the transport demand. Railroads meet 30% of the transport demand.	Bicycling and walking meet 10% of the transport demand
Bioethanol covers about 5% of the passenger transport.	No bioethanol is needed for transportation.
Half of the good transport is diverted to ships and trains.	

Measures for energy demand: Land Transport 2/2

Effects	Policy Change
Reduction of fuel consumption and of primary energy supply from fossil fuel.	Introduction of road pricing and better physical urban planning. Expansion of railroads (higher capacity and higher speed) and implementation of better physical urban planning. Installation of loading stations; electric vehicles are also needed as storage system to accommodate electricity fluctuations.

Measures for energy demand: Aviation

2030	2050	Effects	Policy Change
Domestic flights are reduced to 5% of the current demand.	Same as 2030	Reduction of fuel consumption and of primary energy supply from fossil fuel.	Expansion of railroads and improvement of fuel-combustion technology of the planes.

Measures for energy demand: Water Transport

2030	2050	Effects	Policy Change
Ships will lower the fuel demand by 40% compared to 2008.	Ships will lower the fuel demand by 60% compared to 2008.	Reduction of fuel consumption and of primary energy supply from fossil fuel.	Improvement of fuel-combustion technology of the ships.

Technology for Biomass to Energy conversion

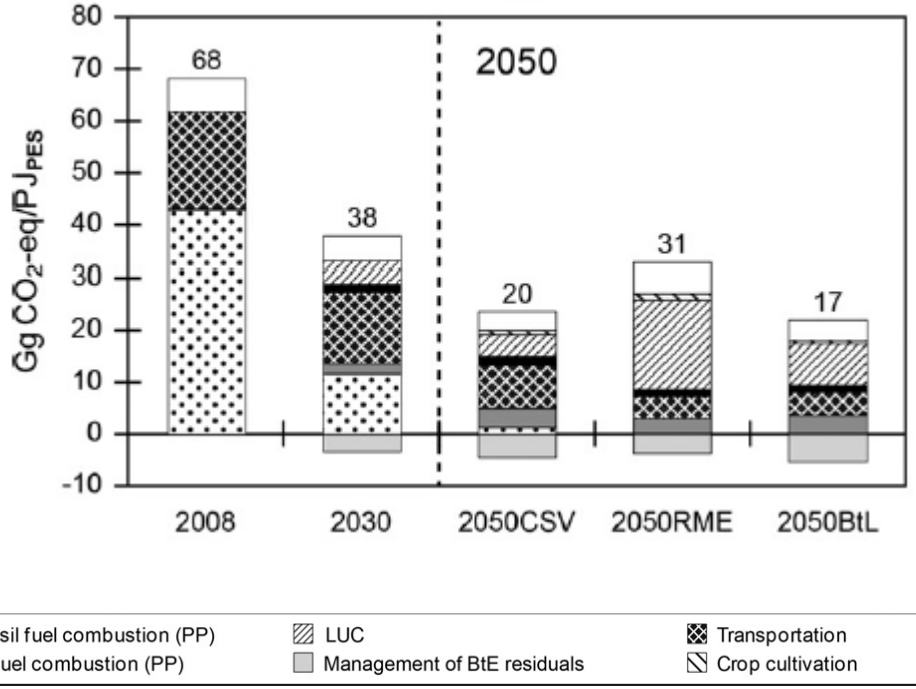
Biomass	Energy technology	Products	Use of products and byproducts
Manure	Anaerobic digestion	Biogas and digestate	Biogas to heat and electricity. Digestate to use on land
Grass	Anaerobic digestion	Biogas, solid biofuel and proteins	Biogas to heat and electricity. Grass fibers to heat and electricity. Proteins substitute soymeal
Wood and willow	Gasification	Syngas and biochar	Syngas to heat and electricity. Biochar to use on land
Straw	Gasification	Syngas and biochar	Syngas to heat and electricity. Biochar to use on land
Waste	Incineration	Electricity and heat	–
Rapeseed	Transesterification	RME, glycerin and solid biofuel	RME to transport. Glycerin substitutes glycerin production. Biofuel to heat and electricity
Willow (BtL, FT)	Gasification and Fischer–Tropsch	FT-biodiesel and biochar	FT-biodiesel for transport. Biochar to use on land
Straw (BE)	Straw refinery	BE, molasses and solid biofuel	Bioethanol for transport. Molasses substitutes fodder. Biofuel to heat and electricity

Comparative energy consumption for transportation sector

Transport	2030							2050			
	E	BD	DS	PE	BE	CH_4/SF	AF	E	BD	CH_4/SF	AF
Passenger cars	10.7	1.3	5.7	27.5	4.7	–	–	17.1	–	5.1	–
Vans	5.9	3.2	18.5	–	–	–	–	7.4	–	2.2	–
Buses	–	–	9.1	–	–	0.8	–	0.9	5.7	1.1	–
Lorries	–	–	33.6	–	–	–	–	–	25.2	3.4	–
Passenger trains	6.6	–	0.2	–	–	–	–	16.9	–	–	–
Freight trains	0.9	–		–	–	–	–	1	–	–	–
Domestic aviation	–	–	–	–	–	–	0.1	–	–	–	0.1
International aviation	–	–	–	–	–	–	37.8	–	–	–	33.4
Ship transport	–	–	2.8	–	–	–	–	–	1.8	–	–
Defence	–	–	2.1	–	–	–	–	–	2.1	–	–
Total	24.1	4.5	72	27.5	4.7	0.8	37.9	43.3	34.8	11.8	33.5

E (electricity), BD (biodiesel), DS (diesel), PE (petrol), bioethanol (BE), synthetic fuel (SF), methane (CH4) and aviation fuel (AF).

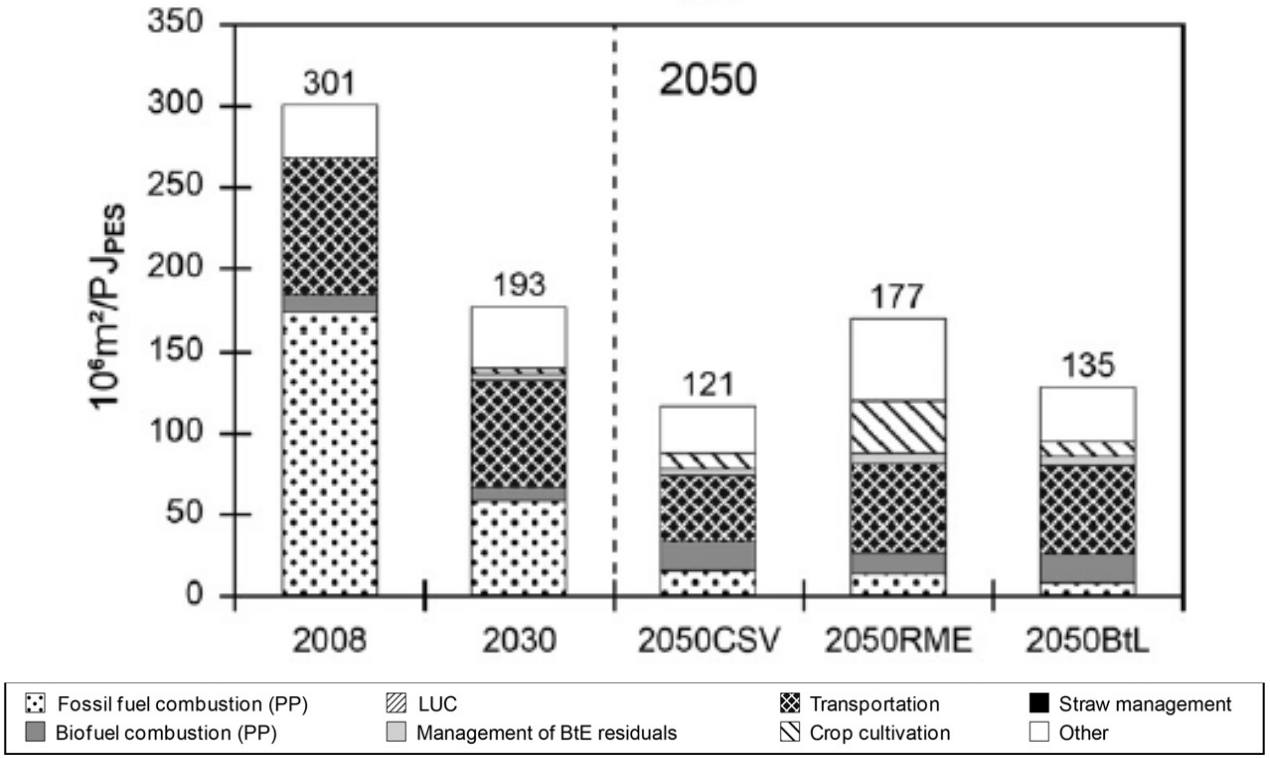
Impact: Global Warming (GW)



Contribution of the different sub-processes to the impact on the selected environmental categories.

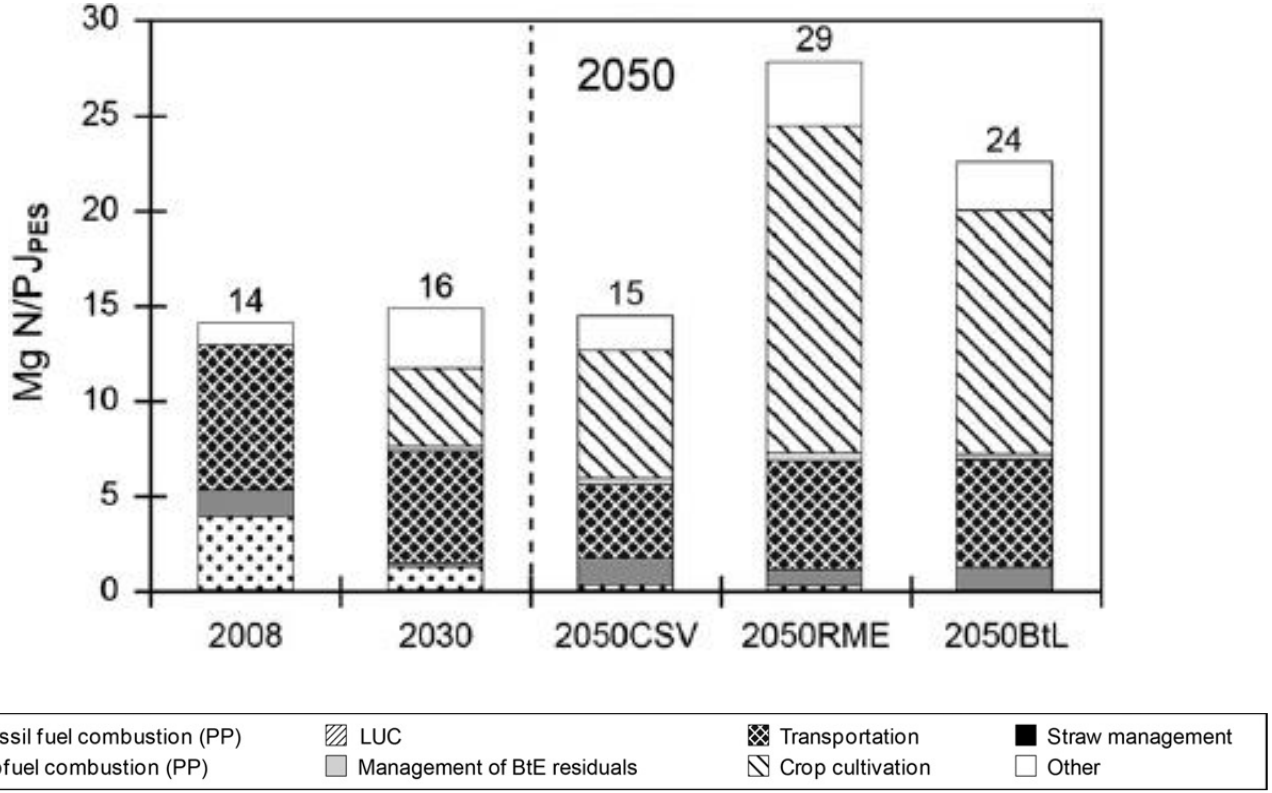
PP: power plants; LUC: sum of direct Land Use Change (dLUC) and indirect Land Use Change (iLUC); BtE: Biomass-to-Energy.

Impact: Acidification (AC)



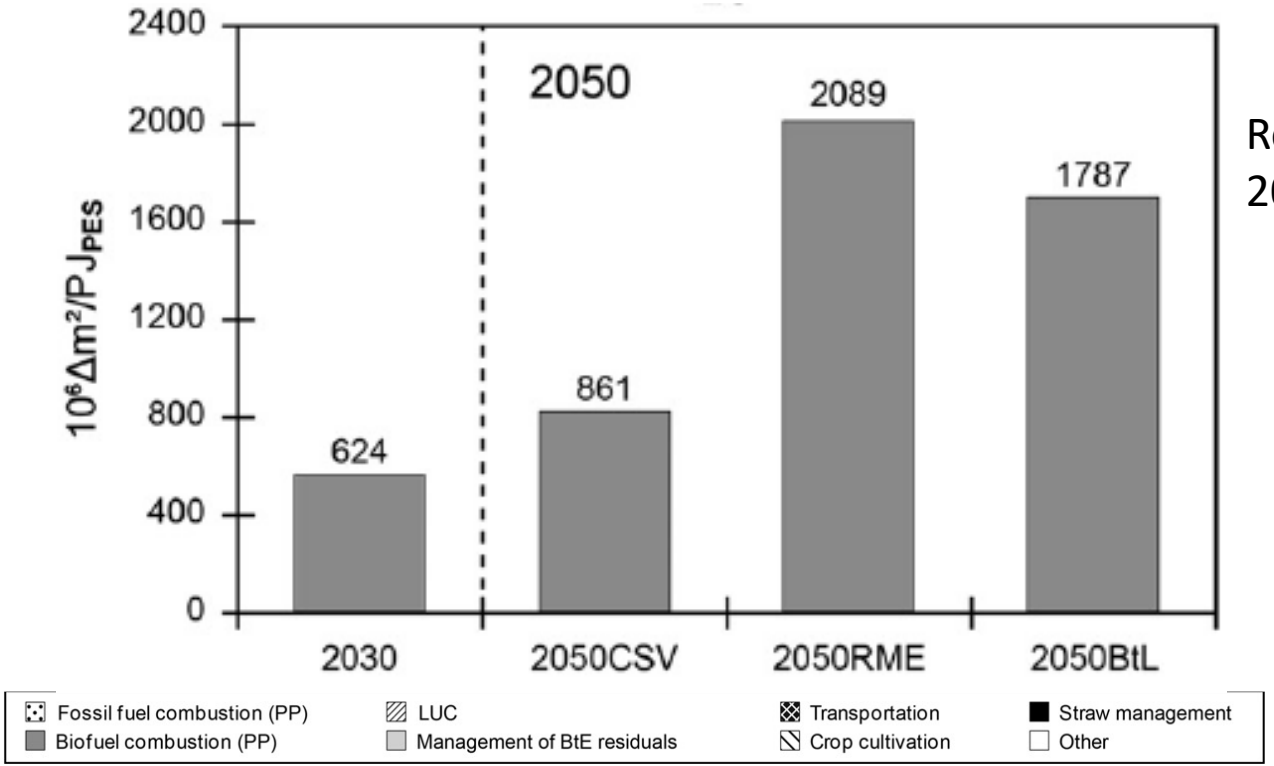
PP: power plants; LUC: sum of dLUC and iLUC; BtE: Biomass-to-Energy; CSV: Conservative Scenario with fossil fuels

Impact: Eutrophication-Nitrogen (EP-N)



PP: power plants; LUC: sum of dLUC and iLUC; BtE: Biomass-to-Energy.

Impact: Additional Land Occupation (LO)



Results are compared to 2008 base scenario

Study Conclusions 1/3

- Significant reductions in GHGs emissions and global warming impacts can be achieved per PJ of energy supplied
- Residual domestic biomass resources were insufficient to cover demand for biomass energy thereby requiring additional cultivation of energy crops which caused significant environmental loads in most impact categories
- Large impacts associated with upstream land use changes (LUCs) made the use of fossil diesel for heavy transport favorable to RME- and FT-biodiesel

Study Conclusions 2/3

- High potential aquatic eutrophication effects were a direct consequence of energy crops cultivation.
- Reduction of the energy demand, increased share of wind power and replacement of fossil fuels with residual domestically available biomasses represented the main means for GHGs emissions savings in the future energy scenarios.
- Main environmental challenge is the supply of biofuels for heavy terrestrial transport, ships, defense and aviation.

Study Conclusions 3/3

- Use of energy crops to fulfill this demand caused significant environmental impacts related to global warming (mainly due to LUC), aquatic eutrophication (increased fertilizers use) and land occupation.

- Use of fossil diesel for these applications appeared preferable over the biomass-based fuels, except for global warming where FT-biodiesel performed slightly better

Conclusions

- LCA is a tool which gives a clear objective understanding of impacts of choices
- LCA is a strong tool to assess the comparative impacts of policy alternative on environment.
- LCA can be widely used in transport policy & planning.

References

- Davide Tonini, Thomas Astrup, “LCA of biomass-based energy systems: A case study for Denmark”, Applied Energy, Volume 99, 2012, Pages 234-246, ISSN 0306-2619

Thank You